

Additional Mathematics

Chapter 7: Coordinate Geometry

NOTES

7.1 Parallel Lines

① Expression of straight line
 $y - y_1 = m(x - x_1)$

② Angle of inclination, θ
 $\tan \theta = \text{gradient}$

③ Gradients of parallel lines
 same gradient (m)

④ Collinear points
 Points that lie on a straight line

7.2 Perpendicular Lines

⑤ Gradients of perpendicular lines
 $m_1 \times m_2 = -1$

7.3 Midpoint of a Line Segment

⑥ Midpoint Formula
 $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$

7.4 Areas of Triangles and Quadrilaterals

⑦ Area of Triangle

$$\text{Area of } \Delta = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{vmatrix}$$

↘ Positive ↗ Negative

⑧ Area of Polygon

$$\text{Area of Polygon} = \frac{1}{2} \begin{vmatrix} x_1 & x_2 & x_3 & \dots & x_n & x_1 \\ y_1 & y_2 & y_3 & \dots & y_n & y_1 \end{vmatrix}$$

↘ Positive ↗ Negative

OTHER FORMULAS TO KNOW

Gradient

$$\text{gradient} = \frac{y_2 - y_1}{x_2 - x_1}$$

Length of line segment

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

QUESTION TYPES

7.1 Parallel Lines

1. Find angle of inclination of two points
 Step 1: Find gradient
 Step 2: Find angle of inclination ②

2. Find equation of line given angle of inclination + one point
 Step 1: Find gradient ②
 Step 2: Equation of line ①

3. Find equation of line parallel to another line given point
 Step 1: Find gradient ③
 Step 2: Equation of line ①

4. Find coordinates of points given that three points are collinear.
 Step 1: Create Equation:
 $\text{Gradient of AB} = \text{Gradient of AC}$
 Step 2: Solve for coordinate.

5. Show/determine whether points are collinear.

Step 1: Find gradients:

Gradient of AB =

Gradient of AC = (Pick two)

Gradient of BC =

Step 2: Explanation to explain whether collinear.

Case 1: Collinear

Since the gradients of AB and BC are equal and B is a common point, A, B and C are collinear.

Case 2: Not collinear

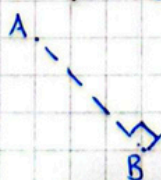
Since gradient of AB $\neq$ gradient of BC, points A, B and C are not collinear.

7.2 Perpendicular Lines

6. Find unknowns in coordinates of points given that lines are perpendicular.

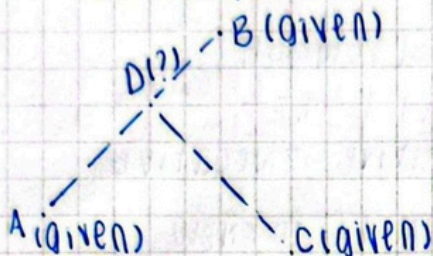
Step 1: Gradients of AB and BC

Step 2: Find coordinates by substituting gradient of AB and gradient of BC into $m_1 \times m_2 = -1$ ⑤



7. Given that the line joining two given points is perpendicular to a line passing through one given point, find point where the lines intersect.

Step 1: Gradient of AB
 Step 2: Gradient of CD using ⑤
 Step 3: Equation of AB and CD ①
 Step 4: Simultaneous equations of eqn of AB and eqn of CD to find point D.

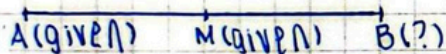


7.3 Midpoint of a Line segment

8. Given midpoint and one point, find the other point.

Step 1: Let coordinates of B be (x, y) .

Step 2: Equate midpoint with midpoint formula. ⑥



9. Find 4th vertex of a parallelogram

Parallelogram Property:

Midpoint of diagonal AC = Midpoint of diagonal BD

Step 1: Let the midpoint of diagonal with known points be M and find M. ⑥

Step 2: Let coordinates of D be (x, y) .

Step 3: Use parallelogram property and equate to midpoint of BD. Since ABCD is a parallelogram M is also the midpoint of BD. ① [8]

10. Determine if four points make a square

Step 1: Check if two adjacent sides are the same length.

Step 2: Find gradient of the two sides and see if their gradients multiply to -1.

11. Find equation of perpendicular bisectors

Step 1: Gradient of AB

Step 2: Midpoint of AB ⑥

Step 3: Gradient of $\perp$ bisector ⑤

Step 4: Equation of $\perp$ bisector using midpoint of AB ①

12. Find point equidistant from 2 points and lie on a line

Step 1: Find $\perp$ bisector of 2 points [1]

Step 2: Solve simultaneous equations of (1) $\perp$ bisector and (2) line to find coordinates of point.

7.4 Areas of Triangles and Quadrilaterals

13. Given two vertices of a triangle, one coordinate of third vertex and area, find other coordinate of third vertex (possible values)

Step 1: Use formula and equate with given area, simplify.

Step 2: Multiply 2 on both sides to get rid of $\frac{1}{2}$.

Step 3: TWO SOLUTIONS:

$$|x| = 4$$

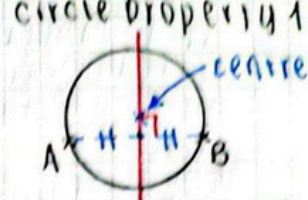
$$\swarrow \quad \searrow$$

$$x = 4 \text{ OR } -x = 4$$

Additional Mathematics

Chapter 8: Circles

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NOTES

8.1 Equation of a circle

① Standard form of circle

$$(x-a)^2 + (y-b)^2 = r^2$$

centre (a, b) and radius r

② General form of circle

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

centre $(-g, -f)$ and
radius $\sqrt{g^2 + f^2 - c}$

③ Coordinates of centre and radius from general form

Centre of circle, (p, q)

$$= \left(\frac{\text{coeff of } x}{-2}, \frac{\text{coeff of } y}{-2} \right)$$

$$\text{Radius} = \sqrt{p^2 + q^2 - c}$$

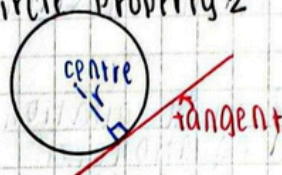
④ Finding equation of circle

1. centre of circle
2. radius of circle

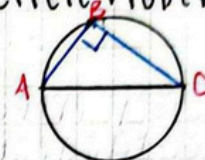
⑤ Circle Property 1: $\perp$ bisector of chord
For any 2 points on the circle,
the $\perp$ bisector of the 2 points always
passes through centre of circle.

⑥ Circle Property 2: Tangent $\perp$ Radius
Any tangent to the circle is always
perpendicular to the radius of the
circle.

Circle Property 2



Circle Property 3



⑦ Circle Property 3: Right angle in a
semicircle

For any triangle in a semicircle where
the diameter is the hypotenuse, it is
a right-angled triangle.

→ Using $\perp$ property (Chapter 7),
formulate an equation and simplify
to find equation of circle.

QUESTION TYPES

8.1 Equation of a circle

1. Find equation of circle given centre and a point

Step 1: Find radius, length of
centre to point.

Step 2: standard form - substitute
values. ①

2. Determine whether points are inside or outside circle.

Step 1: Find radius of circle

Step 2: Use length of line segment
to find length of point
from centre.

Step 3: If length < radius,
inside circle;
If length > radius,
outside circle.

3. Circle Property 1: Find equation of circle given two points and a line which centre lies on. ⑤

Step 1: Find equation of $\perp$ bisector of
two points.

Step 2: solve simultaneous equations
of $\perp$ bisector and line to
get coordinates of centre.

Step 3: Find radius, length of centre to
a point

Step 4: Standard form - substitute
values. ①

4. Circle Property 1: Find equation of circle given three points. ⑤

Step 1: Find equation of $\perp$ bisector
of AB and $\perp$ bisector of BC

Step 2: The centre lies on the
perpendicular bisector of AB
and perpendicular bisector
of BC. (solves simultaneous eqns)

Step 3: Find radius, length of centre
to a point

Step 4: Standard form - substitute
values. ①

5. Circle Property 2. Find the point where tangent touches circle given equation of tangent and centre of circle.

Step 1: Gradient of radius (perpendicular lines)

Step 2: Equation of radius using coordinates of centre.

Step 3: Solve simultaneous equations equation of radius and equation of tangent

5- Extension: Find equation of circle

Step 4: Find radius, length of centre to the point

Step 5: Standard form - substitute values

6. Circle Property 2: Find equation of tangent given equation of circle and a point on circle.

Step 1: Find centre of circle

Step 2: The line joining centre and (point) is a radius. Find gradient of this radius.

Step 3: Since the tangent at (point) is perpendicular to this radius. Gradient of tangent = ? (perpendicular lines)

Step 4: Equation of tangent - substitute coordinates of point

7. Circle Property 3: Find equation of circle given 2 points that make up the diameter.

Step 1: Let a point be (x, y) that lies on circle.

Step 2: Gradient of AB $\times$ gradient of BC = -1 (Substitute coordinates)

Step 3: Simplify the equation to get equation of circle in general form.

8. Find equation of another circle that is a reflection of a given circle in a line.

Step 1: Find coordinates where original circle touches line of reflection (chapter 2 - simultaneous equation to find point of tangent)

Step 2: Use midpoint formula to find centre of other circle. (centre of original circle is one point and midpoint is point where line touches circle)